

WeatherGPT — Applications & Architectural Manual

Comprehensive Guide: Real-World Uses, Standard Weather App Comparison & Full Frontend/Backend Mechanics

1. Real-World Applications & Key Uses of WeatherGPT

WeatherGPT is an AI-powered meteorological decision support platform designed for a diverse spectrum of users. Unlike standard apps that merely display numerical temperatures, WeatherGPT delivers grounded explanations, visual cloud diagnostics, and voice guidance across multiple domains:

- **■ Agricultural Farmers (Kisan Advisor):** Provides crop-specific irrigation advice, fungal pest risk alerts based on relative humidity (>70%), and optimal dry harvest windows.
- **■ Coastal Fishermen & Mariners:** Delivers real-time wave height, sea surface temperature, swell period, and high-tide coastal warnings to prevent accidents.
- **■ Emergency Response & Disaster Management:** Features geofenced alerts for IMD Doppler radar flash floods, severe thunderstorms, and one-click WhatsApp/SMS emergency broadcasts.
- **■ Aviation Pilots & Transport Operators:** Generates METAR airport observations, TAF forecasts, cloud base altitudes, visibility km, and wind shear hazard warnings.
- **■ Rural Citizens & Non-English Speakers:** Eliminates literacy barriers using native voice synthesis (TTS) and speech recognition (STT) across 13 Indian regional languages.
- **■ Everyday Travelers & Citizens:** Offers travel clothing recommendations, UV broad-spectrum sunscreen alerts, and auto-saves the user's permanent home location (e.g. Sathyamangalam).

2. WeatherGPT vs. Normal Standard Weather Apps

The table below compares standard commercial weather applications against WeatherGPT's innovative features:

Feature / Capability	Standard Normal Weather App	WeatherGPT AI Platform
Intelligence & Explanations	Shows raw static numbers (e.g., 32°C) without explaining why or what is happening.	Integrated Google Cloud Gemini 1.5 Flash AI explains atmospheric causes, monsoons, cyclones, and safety steps.
Language & Voice Support	English only or basic machine translation. No native Indian voice speech.	Native 13 Indian Languages Speech Engine (STT/TTS). Prompts user before starting and reads responses aloud.
Visual Photo Analysis	No ability to upload or analyze photographs.	Multimodal Vision AI: Upload or snap cloud/sky/doppler radar photos for AI storm diagnostic reports.
Location Typos & Spelling	Fails or shows 'City Not Found' when typing minor typos (e.g., <i>sathymagalam</i>).	Levenshtein Fuzzy Distance Matching maps typos like <i>sathymagalam</i> → Sathyamangalam (11.504°N, 77.240°E).
Permanent Location Memory	Defaults to GPS location every load or asks user to pick manually repeatedly.	Auto-saves Permanent Home Location (e.g. Sathyamangalam) and reloads it automatically on all future visits.
Emergency Broadcast Sharing	Basic push notification or no emergency sharing buttons.	One-click WhatsApp & SMS Disaster Alert Sharing for coastal & flash flood emergency warnings.
Specialized Industry Portals	Single general forecast page for all users.	Dedicated portals: Farm Advisor, Air Quality (AQI), Marine/Aviation, Flood Risk AI Analyzer, Climate Trends.

1. **Vercel (Frontend)**: Import GitHub repository `PRANAV-26-mv/weatherapp` → Select root directory → Vercel auto-detects `vercel.json` → Click **Deploy**.

2. **Render.com (Backend)**: Create Web Service → Connect `PRANAV-26-mv/weatherapp` → Set Root Directory `backend` → Build Command `pip install -r requirements.txt` → Start Command `uvicorn app.main:app --host 0.0.0.0 --port \$PORT`.